Task

Task: Create 2 EC2 instances on 2 different regions and install nginx using terraform script.

1. **Creating provider.tf file**

ubuntu@ip-172-31-2-193: ~

```
# Mumbai Region Provider
provider "aws" {
  alias  = "mumbai"
  region = "ap-south-1"
}

# Virginia Region Provider
provider "aws" {
  alias  = "virginia"
  region = "us-east-1"
}
~
~
~
```

2. **Creating variable.tf file**

ubuntu@ip-172-31-2-193: ~

```
variable "instance_type" {
  default = "t2.micro"
}

variable "key_name" {
  description = "Your AWS key pair name"
}
~
~
~
~
```

3. **Creating main.tf file**

ubuntu@ip-172-31-2-193: ~

```
data "aws_ami" "ubuntu_mumbai" {
  provider = aws.mumbai
  most_recent = true

  owners = ["099720109477"] # Canonical

  filter {
    name   = "name"
    values = ["ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-*"]
  }
}

data "aws_ami" "ubuntu_virginia" {
  provider = aws.virginia
  most_recent = true

  owners = ["099720109477"]

  filter {
    name   = "name"
    values = ["ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-*"]
  }
}
-- INSERT --                                        8,2           All
```

4. **Creating security groups as security-group.tf file**

ubuntu@ip-172-31-2-193: ~

```
resource "aws_security_group" "sg_mumbai" {
  provider = aws.mumbai
  name     = "nginx-sg-mumbai"

  ingress {
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    from_port   = 80
    to_port     = 80
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
}
~
~
~
~
~
```

ubuntu@ip-172-31-2-193: ~

```
resource "aws_security_group" "sg_virginia" {
  provider = aws.virginia
  name     = "nginx-sg-virginia"

  ingress {
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    from_port   = 80
    to_port     = 80
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
}
~
~
~
~
```

5. **Creating ec2 instances with nginx as ec2.tf file**

ubuntu@ip-172-31-2-193: ~

```
resource "aws_instance" "ec2_mumbai" {
  provider        = aws.mumbai
  ami             = data.aws_ami.ubuntu_mumbai.id
  instance_type   = var.instance_type
  key_name        = var.key_name
  security_groups = [aws_security_group.sg_mumbai.name]

  user_data = <<-EOF
              #!/bin/bash
              apt update -y
              apt install nginx -y
              systemctl start nginx
              systemctl enable nginx
              echo "Hello from Mumbai Region" > /var/www/html/index.html
              EOF

  tags = {
    Name = "Mumbai-Nginx-Server"
  }
}
~
~
~
~
~
~
~
~
```

ubuntu@ip-172-31-2-193: ~

```
resource "aws_instance" "ec2_virginia" {
  provider        = aws.virginia
  ami             = data.aws_ami.ubuntu_virginia.id
  instance_type   = var.instance_type
  key_name        = var.key_name
  security_groups = [aws_security_group.sg_virginia.name]

  user_data = <<-EOF
              #!/bin/bash
              apt update -y
              apt install nginx -y
              systemctl start nginx
              systemctl enable nginx
              echo "Hello from Virginia Region" > /var/www/html/index.html
              EOF

  tags = {
    Name = "Virginia-Nginx-Server"
  }
}
~
~
```

6. **Applying terraform init,validate,plan,apply**

```
ubuntu@ip-172-31-2-193:~$ terraform init
terraform plan
terraform apply
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.34.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
var.key_name
  terraform-key

  Enter a value: terraform-key

data.aws_ami.ubuntu_virginia: Reading...
aws_security_group.sg_virginia: Refreshing state... [id=sg-04b25aeea3c02c259]
data.aws_ami.ubuntu_mumbai: Reading...
aws_security_group.sg_mumbai: Refreshing state... [id=sg-0f0befb6c297cd260]
data.aws_ami.ubuntu_virginia: Read complete after 0s [id=ami-04680790a315cd58d]
data.aws_ami.ubuntu_mumbai: Read complete after 1s [id=ami-07216ac99dc46a187]

Terraform used the selected providers to generate the following execution plan. Resource actions are indic
  + create

Terraform will perform the following actions:

  # aws_instance.ec2_mumbai will be created
  + resource "aws_instance" "ec2_mumbai" {
      + ami                                  = "ami-07216ac99dc46a187"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
```

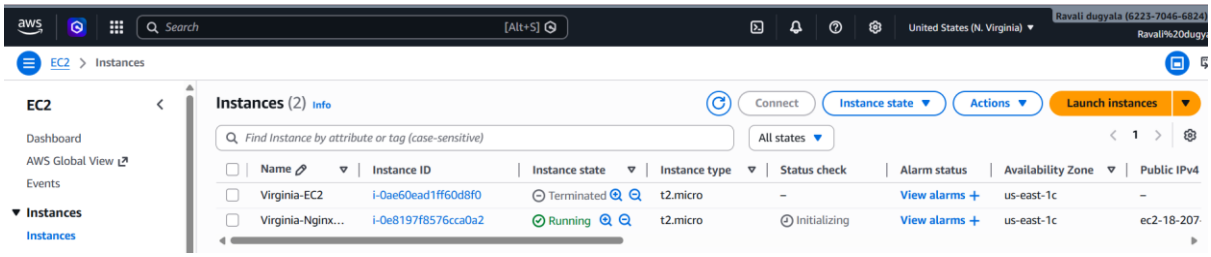
```
  # aws_instance.ec2_virginia will be created
  + resource "aws_instance" "ec2_virginia" {
      + ami                                  = "ami-04680790a315cd58d"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t2.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = "terraform-key"
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + region                               = "us-east-1"
      + secondary_private_ips                = (known after apply)
      + security_groups                      = [
          + "nginx-sg-virginia",
        ]
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
          + "Name" = "Virginia-Nginx-Server"
        }
      + tags_all                             = {
          + "Name" = "Virginia-Nginx-Server"
        }
      + tenancy                              = (known after apply)
      + user_data                            = <<-EOT
            #!/bin/bash
            apt update -y
            apt install nginx -y
            systemctl start nginx
            systemctl enable nginx
            echo "Hello from Virginia Region" > /var/www/html/index.html
        EOT
```

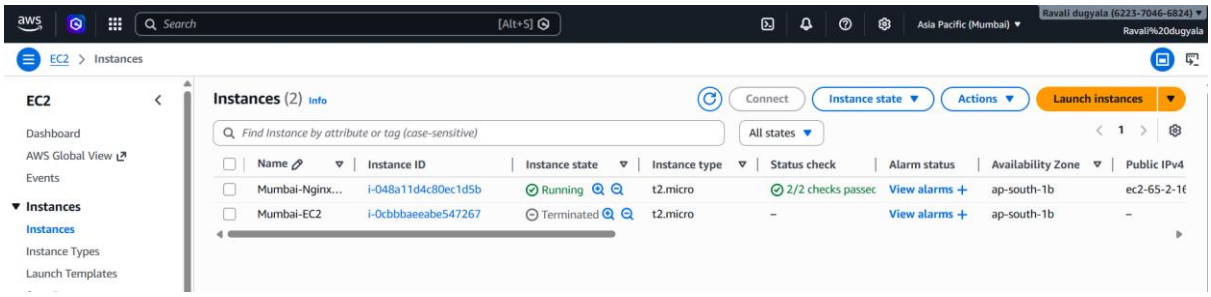
```
Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terrafo
var.key_name
  terraform-key

  Enter a value: terraform-key

data.aws_ami.ubuntu_virginia: Reading...
aws_security_group.sg_virginia: Refreshing state... [id=sg-04b25aeea3c02c259]
data.aws_ami.ubuntu_mumbai: Reading...
aws_security_group.sg_mumbai: Refreshing state... [id=sg-0f0befb6c297cd260]
data.aws_ami.ubuntu_virginia: Read complete after 1s [id=ami-04680790a315cd58d]
data.aws_ami.ubuntu_mumbai: Read complete after 1s [id=ami-07216ac99dc46a187]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symb
  + create

Terraform will perform the following actions:

  # aws_instance.ec2_mumbai will be created
  + resource "aws_instance" "ec2_mumbai" {
      + ami                                  = "ami-07216ac99dc46a187"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t2.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = "terraform-key"
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
```

```
  # aws_instance.ec2_virginia will be created
  + resource "aws_instance" "ec2_virginia" {
      + ami                                  = "ami-04680790a315cd58d"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t2.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = "terraform-key"
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_group_id                   = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
      + public_dns                           = (known after apply)
      + public_ip                            = (known after apply)
      + region                               = "us-east-1"
      + secondary_private_ips                = (known after apply)
      + security_groups                      = [
          + "nginx-sg-virginia",
        ]
      + source_dest_check                    = true
      + spot_instance_request_id             = (known after apply)
      + subnet_id                            = (known after apply)
      + tags                                 = {
          + "Name" = "Virginia-Nginx-Server"
        }
      + tags_all                             = {
          + "Name" = "Virginia-Nginx-Server"
        }
      + tenancy                              = (known after apply)
      + user_data                            = <<-EOT
            #!/bin/bash
            apt update -y
            apt install nginx -y
            systemctl start nginx
            systemctl enable nginx
            echo "Hello from Virginia Region" > /var/www/html/index.html
```

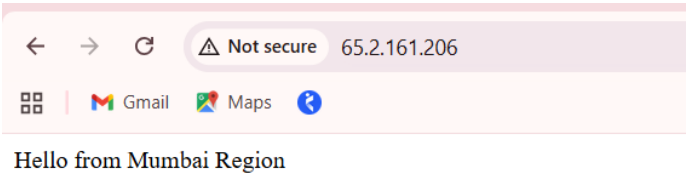
```
    }

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.ec2_virginia: Creating...
aws_instance.ec2_mumbai: Creating...
aws_instance.ec2_virginia: Still creating... [00m10s elapsed]
aws_instance.ec2_mumbai: Still creating... [00m10s elapsed]
aws_instance.ec2_mumbai: Creation complete after 14s [id=i-048a11d4c80ec1d5b]
aws_instance.ec2_virginia: Creation complete after 15s [id=i-0e8197f8576cca0a2]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
ubuntu@ip-172-31-2-193:~$
```

7. Verifying ec2 creation in aws console.

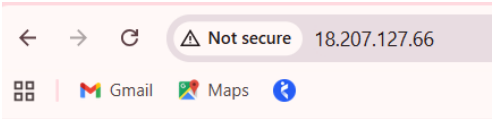




8. Verifying working website in browser.



Hello from Mumbai Region



Hello from Virginia Region